

HACETTEPE UNIVERSITY

Department of Nuclear Engineering

NEM 467 APPLICATION 1

End-of-Cycle Activity of the CAREM-25 Core with KORIGEN

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ABSTRACT

The end-of-cycle activity of the CAREM-25 core was calculated with KORIGEN by editing the VVER-1000 example deck. For 3.1% enriched UO_2 fuel burned to 24 MWd/kg at 26.04 MW/tHM, the total activity is 5.53×10^{18} Bq per tHM and 2.12×10^{19} Bq for the 3.840 tHM core. Of the five variations, the 20 MWd/t burnup lowers Cs-137 by a factor of about 1160 but the total only to 76%, the 20 w/o enrichment lowers Pu-239 to 35%, and the Th-232 case produces U-233 and raises the total by 9%. The U_3Si_2 fuel puts 17% more uranium in the core, so the specific power falls to 22.24 MW/tHM and the short-lived nuclides fall by the same factor. The FeCrAl cladding changes only the activation products.

1 INTRODUCTION

CAREM-25 is a small integral PWR from Argentina with a thermal power of 100 MWt and natural circulation in the primary loop [4,5]. Its core has 61 hexagonal assemblies with 108 fuel rods each. In this application KORIGEN was used to find the total activity of the core at the end of the cycle (EOC) and the activities of fifteen nuclides: U-233, U-235, U-238, Pu-239, I-131, I-132, Cs-137, Cs-134, Sr-90, Ce-144, Ba-137m, Xe-133, Xe-135, Kr-85m and Kr-88 [1]. The same nuclides were then compared for five changes to the base case: a burnup of 20 MWd/t (a), 20 w/o enrichment (b), half of the U-238 replaced by Th-232 (c), U_3Si_2 fuel (d) and FeCrAl cladding (e).

2 METHODS

2.1 CAREM Data

The CAREM data used in the base case are given in Table 1. The core data are taken from [4–6] and the cladding dimensions and pitch from [7]. The gap gas is not given in these sources and was taken as helium.

Table 1: CAREM-25 core and fuel rod data.

Parameter	Value
Thermal power	100 MWt
Number of fuel rods	$61 \times 108 = 6588$
Fuel material	UO_2 (95% of 10.96 g/cm^3)
Enrichment	3.1 w/o U-235
Burnup	24 MWd/kg
Pellet diameter / active length	0.76 cm / 140 cm
Cladding	Zircaloy-4, OD 0.90 cm, thickness 0.0625 cm
Gap material	helium, 75 μm radial gap
Rod pitch	1.38 cm

KORIGEN works per tonne of heavy metal, so the power had to be given in MW/tHM. One rod holds $\pi/4 \times 0.76^2 \times 140 = 63.51$ cm^3 of fuel, and with a uranium fraction of 0.8815 in UO_2 this is $63.51\,\mathrm{cm}^3 \times 10.412\,\mathrm{g/cm}^3 \times 0.8815 = 582.9$ g of uranium. The core then has $6588 \times 582.9\,\mathrm{g} = 3.840$ tHM, and

$$P = \frac{100\,\mathrm{MW}}{3.840\,\mathrm{tHM}} = 26.04\,\mathrm{MW/tHM}, \qquad t = \frac{24\,000\,\mathrm{MWd/t}}{26.04\,\mathrm{MW/t}} = 921.6\,\mathrm{d}. \tag{1}$$

The cladding is 151.0 g per rod, or 259 100 g per tHM, and was split into Zr, Sn, Fe, Cr and Hf with the Zircaloy-4 composition. The steel lines of the VVER deck were kept.

2.2 Input Deck

All runs use the `vver.k.i` deck of the VVER example [2,3]. Structural materials are entered in grams per tHM, actinides in moles per tHM and the irradiation in 18 steps. The VVER cross-section library was kept. The fifth number on line 19 of the deck selects the step from which the decay part starts. It was 10 in the example, which gave the activities at about half of the cycle, so it was changed to 18 in every case. The changes made for each case are listed in Table 2, and the base deck is given in the Appendix.

Table 2: Changes to the base deck (per tHM).

Case	Change
(a)	Irradiation time $20/26.04 = 0.768$ d, all else as in the base case.
(b)	U-235 = 850.905 mol (200 kg), U-238 = 3360.627 mol (800 kg).
(c)	U-238 = 2035.28 mol, Th-232 = 2087.917 mol (484.5 kg each).
(d)	Si = 78 685.4 g, power 22.244 MW/tHM, time 1078.94 d, structure $\times 0.854$.
(e)	Zr, Sn, Hf removed; FeCrAl (75/20/5% Fe/Cr/Al, 7.1 g/cm^3) in the same cladding volume: Fe = 220 360 g, Cr = 58 744.30 g, Al = 14 021.2 g.

Case (a) was run as written, 20 MWd per tonne, which is only 18.4 hours at full power.

Case (d) needed more than adding silicon. U_3Si_2 (12.2 g/cm^3, also taken at 95%) carries 682.4 g of uranium in the same pellet, so the core holds 4.496 tHM. At the same 100 MWt this means $100/4.496 = 22.24$ MW/tHM and a longer time to reach 24 MWd/kg. The cladding and steel per tHM were scaled by $582.9/682.4 = 0.854$. In case (c) the power per tHM was left at 26.04 MW/tHM.

3 RESULTS AND DISCUSSION

The EOC activities are given in Table 3, and Figure 1 shows how each total splits into fission products, actinides and activation products.

Table 3: EOC activities (Bq per tHM).

Nuclide	Base	(a) 20 MWd/t	(b) 20 w/o	(c) Th-232	(d) U_3Si_2	(e) FeCrAl
U-233	1.04×10^{6}	2.13×10^{3}	2.65×10^{6}	4.85×10^{12}	1.05×10^{6}	1.04×10^{6}
U-235	9.88×10^{8}	2.48×10^{9}	1.36×10^{10}	1.10×10^{9}	9.88×10^{8}	9.88×10^{8}
U-238	1.18×10^{10}	1.21×10^{10}	9.92×10^{9}	5.93×10^{9}	1.18×10^{10}	1.18×10^{10}
Pu-239	1.22×10^{13}	2.83×10^{9}	4.22×10^{12}	5.95×10^{12}	1.22×10^{13}	1.22×10^{13}
I-131	2.03×10^{16}	8.80×10^{14}	1.67×10^{16}	2.21×10^{16}	1.73×10^{16}	2.03×10^{16}
I-132	3.66×10^{16}	4.47×10^{15}	3.43×10^{16}	3.65×10^{16}	3.12×10^{16}	3.66×10^{16}
Cs-137	2.82×10^{15}	2.42×10^{12}	2.83×10^{15}	2.86×10^{15}	2.80×10^{15}	2.82×10^{15}
Cs-134	2.97×10^{15}	1.14×10^{8}	5.93×10^{14}	2.47×10^{15}	2.86×10^{15}	2.97×10^{15}
Sr-90	2.07×10^{15}	2.30×10^{12}	2.72×10^{15}	2.59×10^{15}	2.05×10^{15}	2.07×10^{15}
Ce-144	3.24×10^{16}	7.82×10^{13}	3.77×10^{16}	3.40×10^{16}	2.86×10^{16}	3.24×10^{16}
Ba-137m	2.68×10^{15}	3.83×10^{12}	2.68×10^{15}	2.75×10^{15}	2.66×10^{15}	2.68×10^{15}
Xe-133	5.24×10^{16}	1.20×10^{15}	5.29×10^{16}	5.10×10^{16}	4.47×10^{16}	5.24×10^{16}
Xe-135	1.67×10^{16}	1.37×10^{16}	3.79×10^{16}	1.89×10^{16}	1.59×10^{16}	1.67×10^{16}
Kr-85m	6.90×10^{15}	9.75×10^{15}	1.04×10^{16}	1.24×10^{16}	5.90×10^{15}	6.90×10^{15}
Kr-88	1.72×10^{16}	2.68×10^{16}	2.72×10^{16}	3.05×10^{16}	1.47×10^{16}	1.72×10^{16}
Total	5.53×10^{18}	4.22×10^{18}	4.75×10^{18}	6.05×10^{18}	4.73×10^{18}	5.55×10^{18}

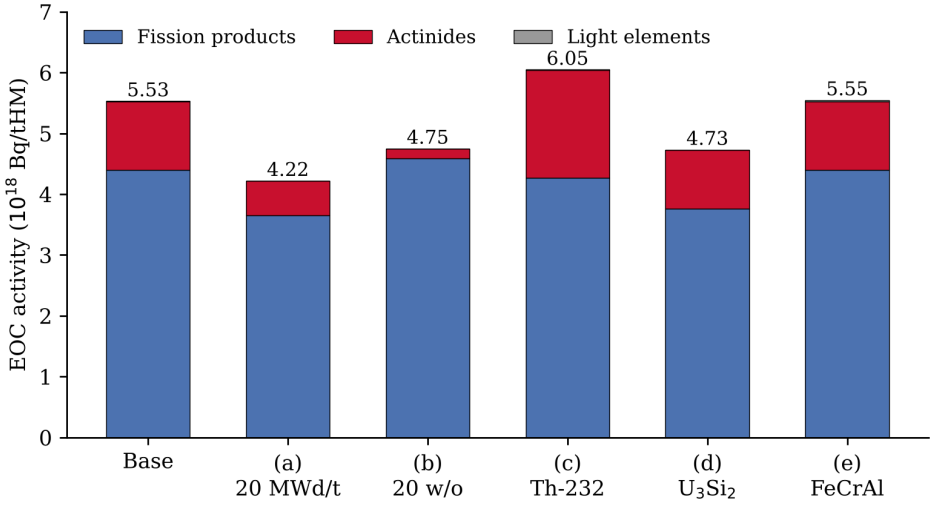


Figure 1: Total EOC activity of each case. The value on each bar is the total in 10^{18} Bq/tHM.

3.1 Base Case

The total is 5.53×10^{18} Bq/tHM. For the whole core this is $5.530 \times 10^{18} \times 3.840 = 2.12 \times 10^{19}$ Bq (5.74×10^{8} Ci). About 80% of it comes from fission products. In the actinides, Np-239 alone is 5.46×10^{17} Bq, almost half of their total. As a check, the decay heat at shutdown is 1.50 MW/tHM, 5.8% of the operating power, which is close to the usual 6%.

3.2 (a) 20 MWd/t

This case used the value in the assignment, 20 MWd/t, reached in 0.768 days at 26.04 MW/tHM. The burnup is therefore 1200 times lower, and Cs-137 and Sr-90 drop by about the same factor. Cs-134 drops much more, to 3.8×10^{-8} of its base value, since it forms by neutron capture in Cs-133 and not directly from fission. The total only falls to 76% because most of the activity at shutdown is from nuclides with half-lives of hours, which are already saturated after 18 hours. Kr-85m and Kr-88 are even higher than in the base case (1.41 and 1.56 times). After 18 hours almost no Pu-239 has formed yet (its activity is 2.83×10^{9} Bq against 1.22×10^{13} Bq in the base case), so nearly all thermal fissions are still in U-235, which gives more of these krypton isotopes than Pu-239.

3.3 (b) 20 w/o

With more U-235 the same power needs less flux (4.03×10^{13} against 2.29×10^{14} n/cm^{2}s). There is also less U-238, so captures in U-238 drop. Pu-239 falls to 0.35 and Cs-134 to 0.20 of the base value, and the actinide total drops from 1.13×10^{18} to 1.66×10^{17} Bq. Xe-135 goes up 2.3 times because the lower flux burns less of it. The total ends up at 86%.

3.4 (c) Th-232

Thorium gives U-233 through Pa-233. U-233 rises from 1.04×10^{6} to 4.85×10^{12} Bq and Pa-233 alone reaches 6.52×10^{17} Bq, which is why the total goes up 9%. U-238 and Pu-239 are halved, as expected from removing half of the U-238. Sr-90 and the Kr isotopes increase because U-233 fission yields are higher on the light side.

3.5 (d) U_3Si_2

The short-lived nuclides drop to 0.854 of the base case, which is exactly the power ratio $22.24/26.04$. Cs-137 and Sr-90 hardly change (0.995 and 0.994) since the burnup is the same. The total per tHM is 86% of the base, but the core holds more uranium, so the core total is $4.731 \times 10^{18} \times 4.496 = 2.13 \times 10^{19}$ Bq, within 0.2% of the base core. That makes sense, since the core power is the same.

3.6 (e) FeCrAl

Fuel, power and burnup are unchanged, so all fifteen nuclides are the same as in the base case. Only the activation products change: they go from 4.6×10^{15} to 2.17×10^{16} Bq, mostly Cr-51 (1.24×10^{16} Bq) and Fe-55, while Zr-95 and Nb-95 disappear. This is still under 0.5% of the total.

4 CONCLUSION

The total EOC activity of CAREM-25 is 5.53×10^{18} Bq per tHM, or 2.12×10^{19} Bq for the 3.840 tHM core, and about 80% of it comes from fission products.

Looking at the cases together, the total at shutdown depends mostly on the power and much less on the burnup. Case (a) ran for only 18.4 hours at 20 MWd/t and still reached 76% of the base total, while case (d) lowered the power per tHM by 15% and the short-lived nuclides dropped by the same amount. The long-lived nuclides behave the other way. Cs-137 and Sr-90 follow the burnup and hardly react to the power. So the activity that matters right after shutdown, mainly iodine and noble gases, is set by the power level, while the inventory that stays in spent fuel for years is set by the burnup.

The enrichment and thorium cases show that the actinides can change a lot while the fission products stay close to the base case. At 20 w/o, Pu-239 fell to 35% of its base value because there was less U-238 and less flux to convert it. With Th-232, U-233 and Pa-233 appeared in amounts that are practically zero in UO_2 fuel, and Pa-233 alone was enough to raise the total by 9%. Spent thorium fuel would therefore need its own treatment in an inventory or safety analysis.

The fuel and cladding materials had the smallest effect on the listed nuclides. FeCrAl left all fifteen of them unchanged and only raised the activation products, which are still under 0.5% of the total. U_3Si_2 changed them only because the same core holds 17% more uranium. For the whole core the short-lived activity stayed within 0.2% of the base, while the long-lived inventory grew with the extra uranium.

All runs use the VVER cross-section library. CAREM has a wider lattice (pitch/OD = 1.53) than the VVER, so its softer spectrum is not represented, and the actinide results, which depend on captures in U-238 and Th-232, are the most likely to change with a CAREM-specific library.

5 REFERENCES

[1] Hacettepe University, Department of Nuclear Engineering. *NEM 467 Application 1*, 2026.

[2] Hacettepe University, Department of Nuclear Engineering. *KORIGEN VVER input descrip-*

tion (course document).

[3] Hacettepe University, Department of Nuclear Engineering. *Running KORIGEN* (course document).

[4] H. Boado Magan, D. F. Delmastro, M. Markiewicz, E. Lopasso, F. Diez, M. Giménez et al. CAREM project status. *Science and Technology of Nuclear Installations*, 2011:140373, 2011.

[5] International Atomic Energy Agency. *Small Modular Reactor Catalogue*, ARIS, 2024. https://aris.iaea.org/Publications/SMR_catalogue_2024.pdf

[6] H. Lestani et al. CAREM 25 fuel cycle optimization with ATFs. IAEA conference paper. https://conferences.iaea.org/event/374/papers/31147/files/12636-CAREMOptimizationWithATFs.pdf

[7] *Nuclear Engineering and Technology*, 2023, Tables 1–2. https://www.sciencedirect.com/science/article/pii/S1738573323005636

6 APPENDIX

Base-case deck. The other decks differ only in the lines given in Table 2.

Carem_temel/vver.k.i

```
VVER nuclear data from ORFI-libraries                                    2  012
     0.701     0.304      2.01     1.-25 0 0 0 0 0 0-1             1
Burn-up averaged cross-sections                60PCU 40  KFKBIB    26         1
     0
   18   18    0    0    0   18    1    0    0    1    0    1
Irradiation 1tHM UOX 3.1% U235 24MWd/kgHM burnup
 26.0412 26.0412 26.0412 26.0412 26.0412 26.0412 26.0412 26.0412 26.0412 26.0412
 26.0412 26.0412 26.0412 26.0412 26.0412 26.0412 26.0412 26.0412
   12.00   65.51  119.01  172.52  226.03  279.54  333.04  386.55  440.06  493.56
  547.07  600.58  654.08  707.59  761.10  814.61  868.11  921.62
1.00000 tHM  initial mass               86400. d
      1.-7     1.-15      1.+0      1.-3      1.-3      1.+0      1.+0
0400000.16     06000017.24    220000143.70   2400002918.48  26000010585.6      5
2800001508.81  400000254225.5 5000003756.90  72000025.91    0     0.0          5
922350131.890  9223804070.560 0     0.0      0     0.0      0     0.0          2
     0     0.0      0     0.0      0     0.0      0     0.0      0     0.0     0
Burn-up dependent cross-sections               60PCU 40  KFKBIB    26    0    1
     0
    0   14    0    0   18    3   -1    0    0    7    0    0
          000       101       303       101       000       000       000
          000       101       303       101       000       000       000
          000       101       303       101       000       000       000
    0
    1
Decay 1tHM UOX 3.1% U235 after 24MWd/kgHM burnup
     0.5      1.     1.5      2.     2.5      3.     3.5      4.     4.5      5.
     5.5      6.     6.5      7.
1 tHM charged to reactor                86400. d    0.00
    0    0    0    0    0    0    0    0    0    0    0    0
```